

# Kaustav Mukherjee

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## EDUCATION

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|---------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| <b>Carnegie Mellon University (CMU)</b><br><i>Master of Science in Computer Vision (GPA 4.22/4.33)</i>                    | Pittsburgh, PA<br>Aug. 2024 – Dec. 2025      |
| <b>National University of Singapore (NUS)</b><br><i>Bachelor of Engineering in Mechanical Engineering (GPA 4.64/5.00)</i> | Singapore, Singapore<br>Aug. 2020 – May 2024 |

## EXPERIENCE

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| <b>AI Engineer</b><br><i>Kaliber Labs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jan. 2026 – Present<br>San Francisco, CA   |
| <ul style="list-style-type: none"><li>Built a real-time multi-view 3D reconstruction pipeline from surgical video using TSDF fusion, driving a 2-day showcase resulting in a signed LOI with a Fortune 500 medical device company and a paper accepted at ECCV</li><li>Developed novel architectures for JEPA-style world models that improve baselines by over 50% on 3D benchmarks</li><li>Improved textual and aural emotion recognition models by architecting a multi-task learning (MTL) framework for categorical classification and regression, driving a 30% absolute jump in accuracy</li><li>Engineered a stateless microservice architecture with NATS and Redis for a multimodal conversational AI PoC</li></ul>                                                                                   |                                            |
| <b>AI Engineer Intern</b><br><i>Kaliber Labs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | May 2025 – Aug. 2025<br>San Francisco, CA  |
| <ul style="list-style-type: none"><li>Architected a modular robotic navigation system with ROS2 and Nav 2, integrating 6+ CV models with C++ behavior trees via shared memory and multithreading, and deployed on the LG Q9 robot and NVIDIA Isaac Sim</li><li>Reduced per-frame edge processing latency from 300ms to 150ms across multiple CV inference tasks</li><li>Led technical integration across 10+ team members and delivered 10+ live demos to investors and stakeholders</li></ul>                                                                                                                                                                                                                                                                                                                  |                                            |
| <b>Software Engineering Intern</b><br><i>Kaliber Labs</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Jan. 2023 – Jul. 2024<br>San Francisco, CA |
| <ul style="list-style-type: none"><li>Accelerated convergence of a surgical camera pose estimation algorithm by 10 times using particle swarm optimization (PSO) to match semantic segmentation masks between the scope view and a digital twin</li><li>Developed an intra-operative camera registration algorithm using Neural Radiance Fields (NeRFs), enabling linear and geodesic measurements between 3D points with above 90% accuracy</li><li>Fine-tuned stable diffusion to generate synthetic training data, expanding surgical tool datasets by up to 5 times</li><li>Employed artificial data augmentation to train multiple semantic segmentation models to above 97% accuracy</li><li>Optimized CV model inference pipelines via ONNX and quantization, reducing memory footprint by 40%</li></ul> |                                            |

## PUBLICATIONS

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| Bychkov, D., Shah, R., Mukherjee, K., So, M. "Surface Coverage Completeness for Quality-Assured Arthroscopic 3D Reconstruction" <i>Proceedings of the ECCV Workshop on DCA-MI, 2026 (To appear)</i>                    |
| Riaz, S., Zheng, H., Roche, A., Zhang, M., Ragu, A. Penachio, S., Mukherjee, K., Jonelagadda, A. "Multi-Task Learning for Non-Canonical Phoneme Recognition via Articulatory Feature Decomposition" <i>arXiv, 2026</i> |

## PROJECTS

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| <b>Bundle Adjustment for 360° Pose Correction</b>   <i>Gaussian Splatting, SfM, SLAM</i>                                                                                                                                                    | Aug. 2025 – Nov. 2025 |
| <ul style="list-style-type: none"><li>Implemented a rig-based Structure from Motion (SfM) pipeline with bundle adjustment for Stella-VSLAM outputs from 360° images, improving Gaussian Splatting reconstruction quality by 10.5%</li></ul> |                       |
| <b>Drone Perception System at CMU Airlab</b>   <i>Python, OpenCV, ROS, Computer Vision</i>                                                                                                                                                  | Aug. 2024 – Nov. 2024 |
| <ul style="list-style-type: none"><li>Designed a depth-aware perception system for a drone with a 2-DOF manipulator using ROS and RGBD cameras</li></ul>                                                                                    |                       |

## SKILLS

**Competencies:** 3D Reconstruction, Computer Vision, SLAM, SfM, NeRF, Gaussian Splatting, Diffusion, World Models, Reinforcement Learning, LLM Fine-tuning, Context Engineering, Model Quantization  
**Frameworks:** PyTorch, Tensorflow, NumPy, OpenCV, ROS2  
**Tools:** Docker, Redis, NATS, ONNX, AWS, Linux, Git  
**Languages:** Python, C++, JavaScript